Muhammad Faizan

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Education

Northern Illinois University Aug 2026 – Present | DeKalb, IL
PhD in Computer Science
- **Research Focus: Reliable AI and Software Engineering**
  - Exploring reliable and trustworthy AI systems at the intersection of artificial intelligence and software engineering, focusing on robustness testing, uncertainty quantification, interpretability, and systematic evaluation of learning-based and generative AI applications.
  - Investigating AI engineering, adversarial testing, failure analysis, and domain-knowledge integration to design and evaluate reliable AI systems, with applications in secure software engineering and AI-driven cybersecurity.

NED University of Engineering & Technology Oct 2020 – Aug 2024 | Karachi, PK
BE Computer Systems Engineering
- **CGPA:** 3.84/4.00 | Passed with distinction **A***

Professional Experience

Northern Illinois University Aug 2026 – Present | DeKalb, IL
Graduate Research Assistant
- Developing autonomous agents to simulate cyberattacks against software systems, with the goal of identifying recurring exploit patterns, attack behaviors, and potential system vulnerabilities.
- Analyzing simulated attack outcomes to develop and evaluate defensive strategies that improve the security, reliability, and resilience of software systems.

Serveden LLC Feb 2026 – Jul 2026 | Fremont, CA
AI Engineer (Remote - Contract)
- Built workflow components extending the AI platform with multi-step AI workflows and integrations, providing functionality similar to n8n for orchestrating AI agents and external tools.
- Developed a coding-agent connector CLI linking users' local environments with Serveden's Cloud Agent, enabling agents to interact with local files, tools, and MCPs.

Data Science Dojo - Ejento AI Sep 2023 – Jan 2026 | Redmond, WA
Software Engineer I - LLMs and AI
- Designed and implemented a ReBAC authorization system using SpiceDB (AuthZed), enabling fine-grained permissions and data lifecycle controls to support GDPR-compliant platform access.
- Built Ejento Builder with LangChain and developed a CodeAct architecture for interactive AI agent creation and dynamic code execution on structured and semi-structured data.
- Integrated Langfuse observability with distributed tracing, prompt hierarchies, and Jinja templates, reducing debugging time by 60% and improving RAG accuracy by mitigating hallucinations.
- Optimized AI infrastructure with Redis semantic caching, reducing LLM latency by 70%, and built an Azure indexing pipeline based on the Saga design pattern supporting 5K+ concurrent uploads across 22 formats with SharePoint/Outlook integration.

UdhaarBook Nov 2022 – Sep 2023 | Karachi, PK
Junior Data Scientist
- Led data warehousing by engineering 30+ AWS Glue–Redshift ETL pipelines processing 1TB+ of MongoDB/RDS data, saving $30K in outsourcing costs and enabling rapid analysis through Redash.
- Built a personalized recommendation system for Saddar.pk using customer and behavioral data, improving product discovery and driving a 30% increase in customer sales.
- Proposed three credit-scoring models for invoice lending, providing data-driven approaches for borrower risk assessment and supporting financial risk research.

Projects

Revisiting Memories - FYDP Aug 2023 – Aug 2024
Multimodal AI-Based Memory Retrieval System
- Developed an episodic memory recall system for individuals with Alzheimer's, memory loss, and visual impairments, enabling natural-language retrieval of previously observed objects and last-seen locations; trained and evaluated YOLOv8, YOLOv10, and YOLOv11 on 15K+ Ego4D frames.
- Built a multimodal memory retrieval system using OpenAI, PostgreSQL, and Streamlit, enabling real-time text, image, and voice-based queries for contextual visual memory and object retrieval.

Indus Virtual Newborn Assistant - NCL Jun 2022 – Jul 2022
Computer Vision Based Anamoly Detection System
- Prototyped a computer vision–based diagnostic tool for Indus Hospital to support early detection of physical abnormalities in newborns, leveraging OpenCV, TensorFlow, and Roboflow to develop a low-cost, scalable solution for neonatal screening and clinical assessment.

Go Cart Jul 2022 – Dec 2022
AI-Powered Cashierless Shopping System
- Developed a computer vision–based autonomous shopping system inspired by Amazon Go, using CNN models to identify and track products during the shopping experience, enabling cashier-less store operations with OpenCV, React Native, and Firebase.

Achievements

Programming and Data Science Competitions
- **1st Place**, ITEC'22, Data Science @ **NEDUET** - among 36 teams
- **1st Place**, Procom'23, Data Science @ **FAST NUCES** - among 35 national teams
- **1st Place**, Spectrum'23, Data Science @ **DSU** - among 13 teams
- **1st Place**, Techfest'24, Speed Programming @ **NEDUET** - among 42 teams

Awards & Honors
- **Internet of Things Developer, PIAIC** - 93rd percentile; recognized by the President of Pakistan
- **Giant Leap Award**, Data Science Dojo - H2 2024
- **Dot Award**, Data Science Dojo - Q4 2025
- **Player of the Quarter**, UdhaarBook - Q2 2023

Volunteer and Teaching Experience

Qubitcoders Nov 2023 – Jul 2024 | Karachi, PK
- Co-founded a 100+ member Competitive Programming Community at NEDUET to raise awareness and promote engagement in problem-solving and contests.

Bootcamp instructor - AI Club Apr 2023 – Jul 2023
- Conducted 12 workshops and mentorship for 80+ undergraduates across Pakistani universities, teaching modules in Python programming, data science, machine learning, and problem-solving.

Section Leader - Stanford CIP 2025 – 2026
- Volunteered as Section Leader for Code in Place, teaching programming fundamentals to 20 students from diverse regions worldwide in the years 2025 and 2026

Certificates

Microsoft Azure
- Azure Solutions Architect Expert
- Azure AI Engineer Associate
- Azure Data Engineer Associate
- Azure Administrator Associate
- Azure Developer Associate

Coursera
- Deep Learning Specialization
- Machine Learning Specialization
- Google Data Analytics